

Title: Adaptive Hybrid RAG Framework for Context-Aware Document Intelligence

Abstract

Organizations and individuals generate and manage a large number of digital documents such as reports, research papers, manuals, policies, contracts, and technical documentation. Retrieving accurate information from these documents is often time-consuming, as traditional keyword-based search systems fail to understand the semantic meaning and contextual relationship between queries and document content. General-purpose Large Language Models (LLMs) can generate natural language responses but may produce hallucinated or inaccurate information when answering questions about domain-specific or private documents that were not part of their training data. This project proposes an Adaptive Hybrid RAG Framework for Context-Aware Document Intelligence that combines Hybrid Retrieval-Augmented Generation (Hybrid RAG) with adaptive retrieval strategies to provide accurate, context-aware, and trustworthy responses from user-uploaded documents.

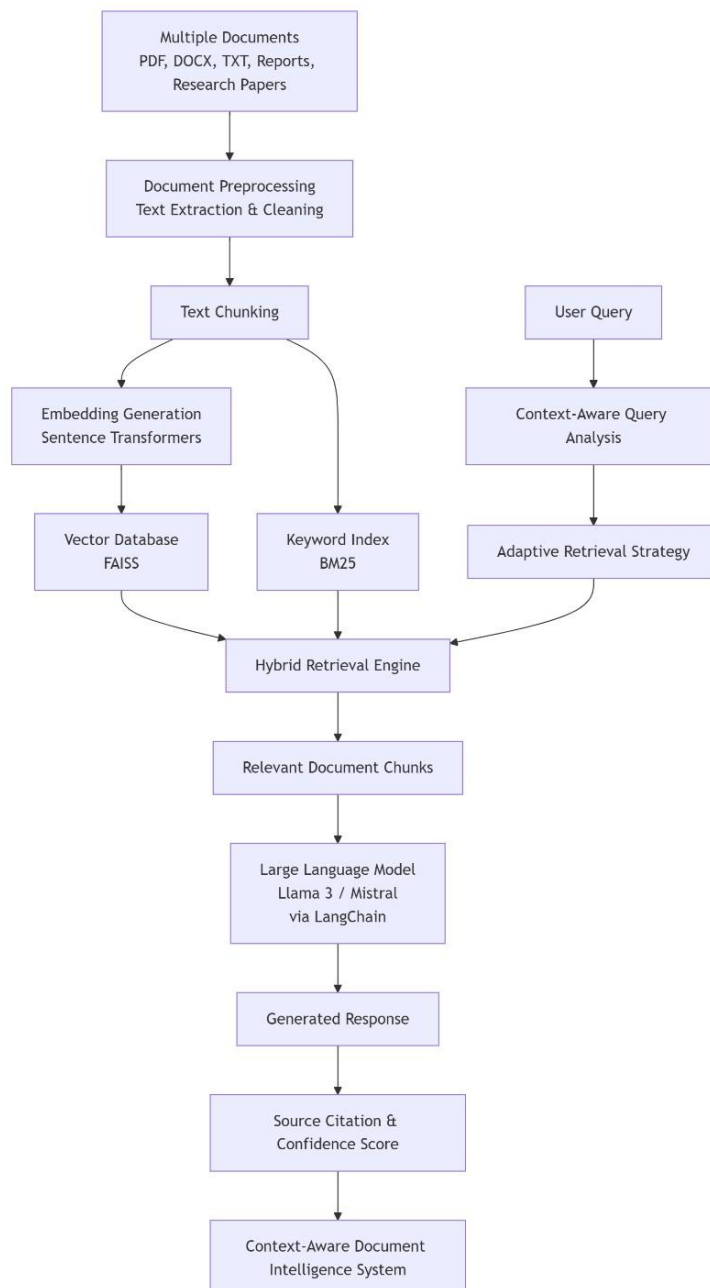
The proposed framework integrates keyword-based retrieval (BM25) and semantic vector retrieval (FAISS with Sentence Transformers) to retrieve the most relevant information based on the user's query. An adaptive retrieval mechanism intelligently selects keyword search, semantic search, or a combination of both depending on the nature of the query. The retrieved document chunks are then provided as contextual information to a Large Language Model (Llama 3 or Mistral) through the LangChain framework, enabling the generation of accurate and human-like responses grounded in the uploaded documents. The system also maintains conversational context to support follow-up questions and provides source citations for every generated answer, thereby improving transparency and reducing hallucinations.

The proposed framework further supports multiple document formats including PDF, DOCX, and TXT files, allowing users to upload and organize large document collections into a searchable knowledge base. Additional functionalities such as document summarization, confidence estimation, context-aware question answering, and conversation history enhance the overall user experience. The performance of the proposed system is evaluated using retrieval metrics such as Precision@K, Recall@K, Mean Reciprocal Rank (MRR), and response quality metrics including answer relevance, faithfulness, latency, and user satisfaction. The resulting framework aims to provide an intelligent document understanding platform capable of delivering reliable, explainable, and context-aware information retrieval for educational institutions, enterprises, research organizations, and other knowledge-intensive domains.

Datasets

Dataset	Purpose	Link
Natural Questions	Open-domain Question Answering	https://huggingface.co/datasets/google-research-datasets/natural_questions
HotpotQA	Multi-hop Question Answering	https://huggingface.co/datasets/hotpotqa/hotpot_qa
MS MARCO	Passage Retrieval & Information Retrieval	https://huggingface.co/datasets/microsoft/ms_marco
BEIR Benchmark	Information Retrieval Evaluation	https://huggingface.co/datasets/BeIR/beir

Approach



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